

Every number, and where it comes from

This is the audit companion to the methodology sheet. That one explains the approach; this one gives the arithmetic. For each figure the dashboard publishes it states the formula, the field it is stored in, the source rows it is built from, and the check that would stop it being published if it were wrong. The last section sets out how to reproduce a figure yourself, from UN Comtrade's own interface, without this code.

How to read this

SECTION 1

Four stages sit between a customs filing and a number on screen. Nothing is computed in the browser: the dashboard reads a stored snapshot in which every figure has already been derived, checked and either approved or withheld.

THE CHAIN		
STAGE	DOES WHAT	CODE
Pull	Requests the aggregate rows and stores them unmodified	pipeline/pull_comtrade.py
Process	Derives every published figure; decides what may be shown	pipeline/process_snapshot.py globaltrade.py, coverage.py
Validate	54 arithmetic and publication checks; blocks on any failure	pipeline/validate_snapshot.py
Publish	12 further checks before the site is allowed to deploy	pipeline/launch_sanity.py

1. How to read this
2. What is requested, and what is discarded — the source rows
3. Whether a period may be published at all — coverage
4. The global trade figure — netting, ranking, share, mirror gap
5. India's own figures — gross, netted, partners, concentration
6. Headings and chapters — and the coverage figure beside them
7. Codes that changed in HS 2022 — lineage and families
8. Tariff lines and rupees
9. HStack — combining codes
10. Rounding, and what a blank means
11. The checks that can stop a number
12. Reproducing a figure yourself

What is requested, and what is discarded

SECTION 2

Every stored row is a reporter's own filing, unmodified. No value is scaled, interpolated, deflated or estimated at any point in the pipeline.

THE REQUEST, PER HS CODE AND PERIOD		
typeCode	C	commodities
freqCode	A / M	annual and monthly
clCode	HS	as filed, in the reporter's own revision
flowCode	M, RM, X, RX	imports, re-imports, exports, re-exports
partnerCode	0	World, for the reporter frame
partner2Code	0	World — all second partners as one
customsCode	C00	TOTAL customs procedure
motCode	0	TOTAL modes of transport

The last three matter more than they look. Left unset, Comtrade returns the second-partner, customs-procedure and mode-of-transport breakdowns *alongside* the aggregate row, and summing an unfiltered response double counts by whatever factor the reporter happens to break its filings down by. Two independent defences apply: the request asks only for the aggregate, and `filter_classic()` discards anything arriving with a non-aggregate value regardless.

A truncated response is refused, not stored

A response returned at the record cap looks identical to a complete one. Any chunk arriving at or above 95% of the cap is rejected and the run stops, because storing it would silently lose rows. `CAP_WARNING_RATIO = 0.95`

Two rows for the same key is a failure, not a sum

Every stored chunk is asserted unique on `cmdCode`, `period`, `reporterCode`, `partnerCode`, `flowCode`. A duplicate would either inflate a world total or silently replace an aggregate with one slice of it.

A failed call stops the run

Three attempts with exponential backoff, every attempt logged to `data/logs/call-errors.jsonl` whether or not a later one succeeded. A call that never succeeds halts the refresh: a snapshot missing a flow would publish an understated world total and look entirely normal doing it.

Whether a period may be published at all

SECTION 3

This test runs before any global figure is derived, and it is the single most consequential rule in the system. UN Comtrade is a rolling collection: a recent period holds whichever economies have filed so far. Summing that as though it were the world understates it, and the shortfall moves month to month — so the dashboard would appear to report a collapse in trade when it was reporting late paperwork.

The test is comparative, not absolute: does this period still contain the economies that mattered in the period before it? Reporters are ranked by value; the previous period supplies the benchmark.

THRESHOLDS – ALL FOUR MUST HOLD		
TEST	VALID	CAUTION
Previous largest reporter present	yes	yes
Of the previous top 10, present	≥ 9	≥ 8
Of the previous top-20 value, retained	≥ 95%	≥ 90%
Reporter count vs previous	≥ 80%	≥ 75%
Anything else		INVALID

`global.coverage.priorTop20ValueCoverage`

= sum of previous top-20 values whose reporter is still present
/ sum of all previous top-20 values

Value-weighted rather than a headcount, so losing one large filer fails the test while losing several small ones does not.

Only VALID publishes

VALID → headline, rank, share and the economy table are published. **CAUTION** and **INVALID** → all four are withheld and the page says which status applied. The reported observations remain in the node and in the workbook export, so an analyst can still see them; nothing on the page implies a completeness the data does not have.

India's filing must reconcile with itself

India appears twice in the source: in its own bilateral return, and as one row inside the all-reporters frame. They describe the same trade. If they differ by more than **max(1% of the direct value, US\$1 million)** the period is demoted to INVALID, because one of the two extracts is stale.

Two statuses mean “not tested”, not “failed”

BASELINE is the first period inside the analysis window, which has no predecessor to compare against. **HISTORICAL** is a period outside that window. Neither publishes a headline. Both are distinguished from INVALID so a blank can be read correctly.

The global trade figure

SECTION 4

One figure, on the import side, net of re-imports, computed economy by economy rather than as a single global deduction.

global.observed.netImports → global.trade

for each reporting economy r :

$$\text{counted}(r) = \begin{cases} M(r) - RM(r) & \text{if } r \text{ filed RM and } RM(r) \leq M(r) \\ M(r) & \text{otherwise} \end{cases}$$

$$\text{global trade} = \sum \text{counted}(r) \quad \text{over all } r$$

M is that economy's imports from World; RM is the re-imports it filed as a separate sub-flow. Published as **global.trade** only when coverage is **VALID**; **observed.netImports** is present either way.

Total imports as filed already contain re-imports — $M = FM + RM + MIP + MOP$ — so a good that left an economy and came back is counted twice unless RM is removed. The removal is per reporter because each files its own sub-flow.

An inconsistent filing is refused, not negated

Where a reporter files $RM > M$, the adjustment is declined for that economy, its gross value is used, and the reporter is recorded in **cIamped**. Subtracting anyway would produce a negative contribution to a world total: arithmetically valid, economically meaningless.

Nothing is imputed for a reporter that does not file RM

Its total is counted as filed. The shortfall is disclosed rather than modelled, which is what the next field is for.

global.observed.adjustmentCoverage

$$= \frac{\sum M(r) \text{ over reporters that filed RM}}{\sum M(r) \text{ over all reporters}}$$

The share of the gross world total that could be adjusted. Read the published figure as **at or slightly below** the true netted value by the unadjusted remainder. Below 25% the QA gate raises a warning against that code and period.

global.indiaShare and global.indiaRank

$$\text{indiaShare} = \text{counted}(\text{India}) / \sum \text{counted}(r)$$

$$\text{indiaRank} = \text{position of India when all reporters are sorted by counted}(r), \text{ descending}$$

Both computed on the **netted** table, not the gross one, so the rank is consistent with the headline. Rank is India's true position among all reporters even when it falls outside the published top ten.

global.mirror.gap

$\text{netExports} = \sum (X(r) - RX(r))$ same rule as above
 $\text{mirror gap} = \text{netImports} / \text{netExports} - 1$

A reconciliation check, not a component of the headline. A modest positive gap is expected: imports are valued CIF and include freight and insurance that the exporter's FOB valuation excludes. Outside the configured band (default 0.70–1.50 on the ratio) it is recorded as a QA warning.

Why the import side is the headline. It matches how import-substitution and PLI analysis is framed, and it has the better coverage — an economy that imports has every incentive to record it at customs, because duty is assessed on it. The export side is computed in full regardless and published as the gap above.

India's own figures

SECTION 5

india.imports / india.exports / india.balance

$\text{imports} = \text{India's reported imports from World}$ (gross)
 $\text{exports} = \text{India's reported exports to World}$ (gross)
 $\text{balance} = \text{exports} - \text{imports}$

Taken from India's own bilateral return, partner = World. These are gross: they are what India filed, and they are what a reader comparing against DGCIS or a ministry release will expect to see.

india.importsNetReImports

= counted(India) from the netting in Section 4

India's own figure net of the re-imports it filed. Carried separately so the headline and the tile can be reconciled to each other, and so the difference between them is visible rather than implied.

india.suppliers / india.destinations

$\text{share(partner)} = \text{value(partner)} / \text{India's gross world total}$
 $\text{coverage} = \sum \text{partner values} / \text{India's gross world total}$

Partner rows are gross, and the denominator is the gross world total. Re-imports are not filed by partner — no authority records that a particular return shipment came back from a particular country — so netting is impossible at this level. Recorded in the data as **basis: "gross"** and stated on the page.

india.suppliers.hhi and top3Share

$$\text{hhi} = \sum (\text{value}(\text{partner}) / \text{world total})^2$$

$$\text{top3Share} = \sum \text{the three largest partner shares}$$

both computed only when $0.95 \leq \text{coverage} \leq 1.05$

Concentration is only meaningful when the bilateral rows actually add up to the reported world total. Where they do not, both are null rather than approximate. Thresholds: ≥ 0.25 high, ≥ 0.15 moderate, below that low.

Headings and chapters

SECTION 6

HS-4 and HS-2 figures are Comtrade's own aggregates, requested at those levels. They are **not** built by summing the six-digit codes this dashboard tracks.

Summing would publish a partial figure under the heading's own name. Of the 106 headings in scope only 30 are fully covered by the sector definition, which holds 418 of the 805 official six-digit lines inside them. Heading 8501 would show one line out of seventeen. It would also destroy the continuous long series, which works precisely because a heading pulled whole contains its own 2022 splits internally.

So each parent page publishes the official figure and, beside it, how large a slice of it the definition tracks:

definitionShare

$$\text{lineShare} = \text{tracked six-digit lines} / \text{official six-digit lines}$$

$$\text{globalShare} = \sum \text{tracked members' global.trade} / \text{the heading's own global.trade}$$

$$\text{indiaImportShare} = \sum \text{tracked members' india.imports} / \text{the heading's own india.imports}$$

Official line counts come from Comtrade's own H6 reference table, stored in `config/reference/` so the count does not depend on network access. Members whose own period failed coverage are omitted from the sum and counted in `membersWithTrade`, so the share is a **lower bound** whenever that number is below `members`; the page says so.

A share above 100% is impossible and is treated as such. The tracked lines are inside the heading, so their sum cannot exceed it. If it does, the heading and its members were built from different data — the QA gate raises a warning and the page states plainly that the figure cannot be right.

Codes that changed in HS 2022

SECTION 7

HS 2022 is the base. A code retired by that revision gets no page of its own; it is recorded as its successors' classification history.

A split is never spliced

851712 covered all cellular handsets until HS 2022 split it into 851713 smartphones and 851714 others. The old total belongs to both successors jointly, and dividing it between them requires a share nobody has published — an invented share would be indistinguishable from measurement. So the six-digit series genuinely begins in 2022, with the predecessor drawn beside it as a separate dashed line that never joins.

Only an identical code may be joined

Where a code was renumbered with no change of scope, the predecessor's years are joined on and the node records `spliced: true`. The QA gate fails any snapshot that sets this on a split or a merge.

Summing a family is legitimate; splitting one is not

`lineage.family` lists the code, its predecessors and its siblings. Before the revision only the predecessor carries data; after it, only the successors. They are **time-disjoint**, so their sum needs no apportionment at all — which is why HStack offers a continuous combined series across the revision while the individual page refuses one.

Filings under a dead code after its revision year are stragglers rather than history — 851712 still shows a trickle — so the predecessor series is truncated at its recorded `validTo` year.

Tariff lines and rupees

SECTION 8

`tariffLines.financialYears[fy]`

`imports/exports` = as filed by DGCIS, per ITC(HS)-8 line
`balance` = `exports` - `imports`
`totals` = Σ over the lines beneath this code
 (null, never 0, when nothing is convertible)

Stored **outside** the calendar-year block. DGCIS reports April–March and Comtrade reports January–December; the QA gate fails any snapshot that files one under the other, and the importer refuses a bare four-digit year rather than guess which financial year it meant.

Currency conversion

`value(INR)` = `value(USD)` × `rate(period, basis)`
`value(USD)` = `value(INR)` ÷ `rate(period, basis)`

`rate`: RBI annual average reference rate
 FY from Economic Survey Statistical Appendix Table 5.4
 CY as the mean of that year's RBI monthly averages

Each period converts at **its own** rate. A single fixed rate across the history would turn an exchange-rate movement into an apparent trade movement. A value filed in rupees is shown exactly as filed and never round-tripped; the other currency is marked derived.

No rate, no conversion

A period with no sourced rate is not converted: the figure stays in dollars and the page says the rate is missing. No interpolation, no nearest-year fallback, no carry-forward. A rate row without a source is refused at load time.

A part year is labelled and excluded from comparison

An export covering fewer than twelve months is marked incomplete, is not chosen as the default view, and is skipped by the reconciliation check — nine months against a full calendar year tells you nothing.

The reconciliation check is a units check

Summed tariff lines are compared against the Comtrade six-digit India figure with a deliberately wide band (0.5–2.0). The two are different statistical systems over different twelve-month windows and will never agree closely. Its job is to catch rupees loaded into the dollar column, which would be roughly 85× out.

HStack

SECTION 9

Combined totals

$\text{combined global trade} = \sum \text{global.trade over codes where it is published}$

$\text{india imports/exports} = \sum \text{india.imports} / \text{india.exports}$

$\text{shareOfBasket}(\text{code}) = \text{code's global.trade} / \text{combined global trade}$

A code whose period failed coverage contributes **nothing** and is listed with the reason. The panel states how many of the stacked codes were counted, so a total is never mistaken for a complete one.

Aggregated economies and suppliers

$\text{value}(\text{partner}) = \sum \text{that partner's value across every stacked code}$

$\text{hhi} = \sum (\text{merged value}(\text{partner}) / \text{total})^2$

HHI is computed on merged per-partner totals. Squaring each product's fragment of a partner's trade separately would report a basket as far less concentrated than it is. The economy ranking is assembled from each product's published top-ten table, so the panel also states what share of basket trade those tables cover.

Overlaps are detected, not silently summed. Stacking a heading together with a product inside it counts that trade twice. HStack finds these pairs and says so at the top of the panel rather than quietly producing an inflated total.

Rounding, and what a blank means

SECTION 10

PRECISION	
Monetary values	Rounded to whole US dollars on storage
Shares, ratios, coverage	Six decimal places
Mirror ratio and gap	Four decimal places
Exchange rates	Four decimal places, as published
Display	Two significant decimals with a magnitude suffix; the stored value is the auditable one

Rounding happens once, at storage. Nothing is rounded and then used in a further calculation, so a displayed total always equals the sum of the displayed parts to within one dollar — which is what the QA gate's arithmetic checks assert.

A BLANK IS NOT A ZERO ON SCREEN		MEANS
—		Not published. Either no reporter filed it, or coverage validation withheld it. The page states which.
0		A reported value of zero.
A break in a line		A year that failed coverage. Lines are never drawn through a withheld period.
Dollars in rupee mode		No sourced rate for that period.

The checks that can stop a number

SECTION 11

54 failure checks and 6 warning checks run over every node before a snapshot may be promoted, and 12 more before the site may deploy. A single failure blocks the entire snapshot; the previously validated one keeps serving.

WHAT IS ASSERTED	
GROUP	EXAMPLES
Arithmetic	balance = exports – imports; net = gross – removed; no negative totals or removals
Bounds	shares within [0, 1]; HHI within [0, 1]; adjustment coverage within [0, 1]; rank ≥ 1
Ordering	ranked tables sorted descending; ranks sequential from 1; shares summing to ≤ 1
Publication	no headline, rank, share or economy table for a period that is not VALID; a VALID period must carry a positive figure equal to net imports
Periods	annual keys are four-digit calendar years; tariff lines carry financial-year labels and never appear inside the calendar-year block
Lineage	no product node for a retired code; a split or merge may never be spliced
Currency	every published rate carries a source and is positive; no derived rupee figure without a sourced rate for that exact period
Coverage	tracked lines cannot outnumber the heading's official lines; coverage shares cannot be negative
Warnings only	mirror gap outside band; adjustment coverage below 25%; tariff-line total far from the six-digit figure; part years; a coverage share above 100%

Every stage fails closed

An incomplete pull, a QA failure, a schema mismatch or a snapshot built from fixture data all stop the process and leave the live snapshot exactly as it was. A stale but validated dashboard is preferable to a fresh but wrong one, and a partial snapshot is the dangerous case — it is missing periods rather than visibly broken.

Reproducing a figure yourself

SECTION 12

Every published figure can be rebuilt from UN Comtrade's own interface without running any of this code. The steps below reproduce a global trade headline; the same approach works for any field.

- 1. Read what the dashboard claims.** Open the product page, note the year, and download the workbook. The Annual sheet carries `grossGlobalImports`, `reImportsRemoved`, `netGlobalImports` and `reImportAdjustmentCoverage` for that year.
- 2. Fetch the same rows.** In Comtrade Plus, request: reporters = All, partner = World, second partner = World, customs procedure = TOTAL, mode of transport = TOTAL, commodity = the HS code, flow = Imports, period = the year. Then repeat with flow = Re-imports.
- 3. Check the gross total.** Sum the Imports rows. It should equal `grossGlobalImports` to within a dollar. If it does not, the likely cause is a second-partner or mode-of-transport breakdown left in the export — the aggregate filters above are what remove it.
- 4. Net it reporter by reporter.** Join the two tables on reporter. Subtract Re-imports from Imports where a reporter appears in both *and* its re-imports do not exceed its imports; otherwise keep the gross value. The sum is `netGlobalImports`, which is the headline.
- 5. Check the disclosure.** The imports of reporters that appeared in the re-imports table, divided by the gross total, is `reImportAdjustmentCoverage`.

6. **Check India.** India's netted value divided by the headline is the published share; its position in the sorted netted table is the published rank.

If a figure is blank, check the status before the arithmetic. The workbook's `globalTradeStatus` column gives the coverage verdict for that year. A blank against CAUTION or INVALID is the system working; the underlying observations are still in the sheet, and re-running the steps above on them will produce a number the dashboard deliberately declined to publish.

Two figures cannot be reproduced from Comtrade alone. Tariff-line detail comes from a DGCIS / TradeStat export, which is committed to the repository and diffable. Rupee conversion uses the rate table in `config/fx_inr_usd.csv`, where every rate carries its source; the rates in force travel with the snapshot under `manifest.currency.rates`, so any rupee figure on the page can be traced to the exact number that produced it.

Source: UN Comtrade, all reporters, partner World, aggregate second partner / customs procedure / mode of transport. India tariff-line detail at ITC(HS) eight digits from DGCIS / TradeStat when supplied, on financial years. Rupee conversion at the RBI annual average reference rate. Official heading line counts from Comtrade's H6 reference table.

Field names in this document are the literal keys in the published snapshot. Every figure on the dashboard can be located in `public/data/snapshots/current/` and traced back through the code paths named in Section 1.